# N200M: Visual Navigation Module (VIO)

# Technical Specifications

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# 1.Overview

The N200M Visual Navigation Module (VIO) is an integrated solution designed for drone environmental perception and positioning needs. It comprises an AI main board, USB camera, and laser rangefinder. The module's core functionality fuses visual, laser, and IMU data from the flight controller to provide drones with real-time, precise positioning navigation, environmental awareness, and assisted obstacle avoidance capabilities. This addresses the limitations of traditional GPS-dependent drones, such as scenarios with GPS interference or weak signals.

Core components and key features include:

(1) AI Main board (Core Computing Unit)

Equipped with a high-performance computing main controller, it processes visual data from the USB camera and inertial data from the flight controller's IMU (Inertial Measurement Unit). Through algorithms, it enables real-time attitude calculation and position localization for the drone.

(2) High-Definition Camera (Visual Perception Unit)

Provides high-resolution visual input, capturing environmental imagery to furnish the algorithm with foundational image data for feature point extraction and motion estimation.

(3) Laser Rangefinder (Auxiliary Perception Unit)

Assists the drone in achieving precise altitude measurement, obstacle distance detection, and terrain following, compensating for the limitations of visual systems in low-light or texture less environments.

# 2.Working Principle

The N200M Visual Navigation Module (VIO) estimates the drone’s motion relative to the ground by observing ground textures through a downward-facing camera, thereby solving the positioning problem when the GPS positioning system is unavailable or has failed.

This module does not require preloaded satellite maps and does not rely on matching with existing maps. It can start operating directly in any unfamiliar area as long as ground textures are present.

The module uses a visible-light camera and a laser rangefinder to observe the ground. Combined with the drone's current attitude, it outputs both positioning data and terrain height to the flight controller via a serial interface. Under normal conditions, the module provides positioning data to the flight controller at a frequency of 30 Hz. However, when operating over textureless or weak-texture areas where valid positioning cannot be obtained, the module will suspend the output of positioning data and automatically resume output once valid texture information is available again.

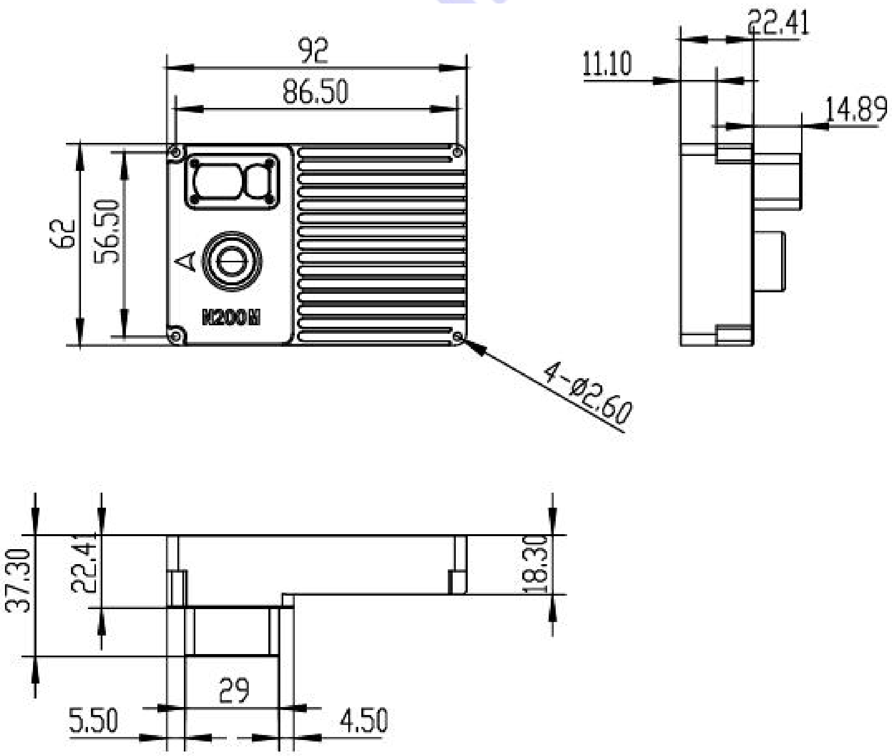
Since heading information must be provided by the magnetometer when GPS is unavailable, magnetometer output directly affects the positioning accuracy of the VIO system. It is recommended to use a high-accuracy, anti-interference external magnetometer to achieve optimal performance.

## 3.Product List

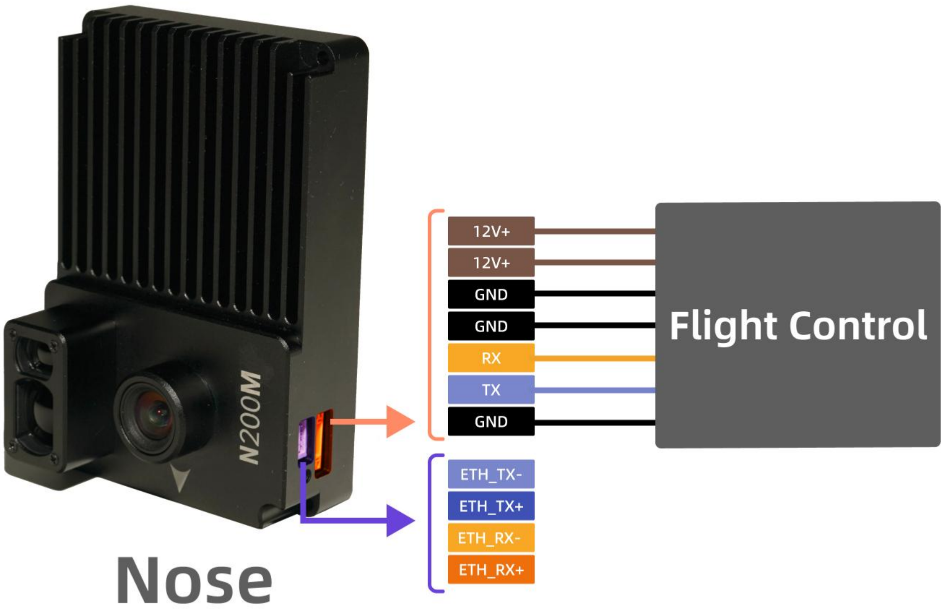
N200M Visual Navigation Module (VIO)*1    Power and Communication Cable*1


## 4.Product Dimensions

Product Dimensions (L×W×H): 92mm×62mm×37.3mm



## 5.Product Interfaces




## 6.Features and Functions

- **Ignore to GPS jamming:**
  - Fuses visual/IMU data for GPS-denied waypoint navigation
  - Executes pre-mapped routes under active EM interference

- **Precise Navigation:**
  - Navigation accuracy: within 5%
  - Optimized for dynamic scenes, this visual sensor eliminates motion blur caused by high-speed movement to achieve precise positioning.

- **Easy to deploy:**
  - Operates without dependency on pre-existing geospatial data
  - Adheres to MAVLink Standard Protocol
  - No development required—just simple configuration

- **Compatibility:**
  - Compatible with a variety of drone models
  - Supports PX4 and APM flight controllers(more platforms coming)

# 7.N200M: Visual Navigation Module (VIO) Technical Specifications

| Category | Parameter | Specification |
|---|---|---|
| General Performance | **Maximum Flight Altitude** | 30~200m |
| | **Maximum Flight Speed** | 15m/s |
| | **Navigation Accuracy** | within 5% |
| | **Output Frequency** | 30Hz |
| | **Supported Flight Controllers** | PX4 / APM |
| | **Applicable Environment** | Suitable for most scenarios (night scenarios not supported yet, to be available soon) |
| | **Output Protocol** | MAVLink |
| Electrical Performance | **Input Voltage** | 12 V |
| | **Communication Interface** | UART TTL 3.3V |
| | **Power Consumption** | 7W |
| Dimensions and Weight | **Dimensions (L×W×H)** | 92mm×62mm×37.3mm |
| | **Weight** | 128g |